



আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية شيتا خونج
International Islamic University Chittagong

Project Report

Project Title: LLM based intrusion detection system

COURSE CODE: CSE - 4744

COURSE TITLE: Computer Security Lab

SUBMITTED TO:

Hasan Shorif
Adjunct Lecturer
Department of CSE, IIUC

SUBMITTED BY:

Kazi Md Safayat Khan - C231181 [8DM]
Ameer Hamzah Daiyan – C231185 [8DM]
Mohammad Moaz – C231187 [8DM]

SUBMISSION DATE:

02-08-2026

REMARK:

LLM-Powered Intrusion Detection System (LLM-IDS) Analysis Report

1. Introduction

The LLM-IDS project is an innovative Intrusion Detection System (IDS) that leverages a local Large Language Model (LLM) to classify network traffic. Its primary goal is to identify network flows as **Benign**, **Suspicious**, or **Attack** with plain-English explanations, incident report generation, providing analyst feedback, all presented on a live Streamlit dashboard. The system is designed for real-time packet capture and analysis, offering a modern approach to network security monitoring.

2. Architecture Overview

The LLM-IDS employs a modular and decoupled architecture, ensuring flexibility and maintainability. The core pipeline involves several stages:

A key design principle is the complete decoupling of the detection pipeline (*main.py*) from the dashboard (*dashboard/app.py*). They communicate solely through a SQLite database (*storage/flows.db*), allowing independent development and future replacement of either component without affecting the other.

3. Core Components

The project is structured into several directories, each housing specific functionalities:

- ***sniffer/***: Handles live packet capture using Scapy, flow tracking (grouping packets into flows based on 5-tuple and timeout logic), and offline PCAP file ingestion.
- ***features/***: Contains the *extractor.py* module, responsible for computing statistical, protocol, and flag-related features from network flows.
- ***analyzer/***: This crucial component integrates the LLM. It includes:
 - *prompt_builder.py*: Formats extracted flow features into a prompt for the local LLM and defines the expected JSON schema for the LLM's response.
 - *llm_client.py*: Manages API calls to the local Ollama LLM for classification, text generation, and JSON generation.

- *query_parser.py*: Translates natural language questions into SQL filters for querying historical flow data.
 - *report_generator.py*: Generates detailed Markdown incident reports from LLM verdicts.
- **storage/**: Manages the SQLite database (*db.py*) for storing classification results and analyst feedback.
- **dashboard/**: The Streamlit web application (*app.py*) that provides a user interface for live capture, PCAP upload, attack simulation, and querying historical data.
- **simulator/**: Contains *traffic_generator.py* for creating synthetic network traffic (e.g., SYN floods, port scans) using Scapy, which is useful for testing and demonstrations without affecting a live network.
- **main.py**: The orchestrator script that runs the sniffing, flow tracking, feature extraction, LLM classification, and storage pipeline.

4. LLM Integration

The LLM plays a central role in the IDS, performing several critical functions:

- **Classification**: The LLM classifies each network flow as **Benign**, **Suspicious**, or **Attack** based on extracted features. It also provides a confidence score and a plain-English explanation for its decision.
- **Incident Reporting**: For classified flows, the LLM can generate a comprehensive Markdown incident report, including a summary, evidence, potential impact, and recommended remediation steps.
- **Natural Language Querying**: Users can ask natural language questions about historical flow data (e.g., "show me attack flows targeting port 22 in the last hour"). The LLM translates these questions into SQL filters, ensuring it never directly executes SQL queries, thus maintaining security.

The system is designed to be fail-safe; if the LLM call fails or returns malformed JSON, the flow is marked as **Suspicious** rather than being silently dropped or classified as **Benign**.

5. Features

The Streamlit dashboard offers a comprehensive interface with four main tabs:

LLM-Powered Intrusion Detection System

Network flow classification via local LLM analysis

[Live Capture](#) [Upload PCAP](#) [Simulate Attacks](#) [Ask](#)

Capture live traffic from a network adapter

Runs the same sniff → flow → feature → LLM pipeline as `main.py`, but started and stopped right here. Capturing raw packets needs admin/root privileges — if this Streamlit process wasn't launched elevated, starting capture will fail with a clear error rather than silently doing nothing.

Network adapter

Microsoft Wi-Fi Direct Virtual Adapter #3

Start Capture

Summary

Total flows

5

Benign

2

Suspicious

3

Attack

0

Flow-by-flow results

flow_id	classification	confidence	explanation
0 192.168.1.253→192.168.1.3:1396/TCP	Suspicious	0.750000	High packet rate (12 packets in 46ms) and SYN imbalance (2 SYN, 11 ACKs) suggest unusual network activity.
1 192.168.1.2:21→192.168.1.3:1399/TCP	Suspicious	0.700000	The flow shows an abrupt connection reset with a high RST count (3) and mild SYN imbalance, indicating unusual beha
2 192.168.1.2:21→192.168.1.3:1402/TCP	Benign	1.000000	The flow shows a balanced request and response traffic with expected TCP handshakes, and reasonable packet counts
3 192.168.1.2:23→192.168.1.3:1403/TCP	Suspicious	0.800000	The flow exhibits one-way communication with few responses (9 packets to the destination vs 7 packets to the source).
4 192.168.1.2:80→192.168.1.3:1404/TCP	Benign	0.900000	Reasonable packet counts and transfer rates, with balanced request and response traffic, indicating normal client serv

Download results as CSV

Drop a .pcap or .pcapng file here

dns-remoteshell.pcap 34.4 KB

Flow idle timeout (seconds)

120

Analyze File

dns-remoteshell.pcap (24.4 KB)

Parsing packets... 131/131

Latest: 192.168.1.2:21→192.168.1.3:1402/TCP → Benign (100% confidence)

flow_id	classification	confidence	explanation
0 192.168.1.253→192.168.1.3:1396/TCP	Suspicious	0.750000	High packet rate (12 packets in 46ms) and SYN imbalance (2 SYN, 11 ACKs) suggest unusual network activity.
1 192.168.1.2:21→192.168.1.3:1399/TCP	Suspicious	0.700000	The flow shows an abrupt connection reset with a high RST count (3) and mild SYN imbalance, indicating unusual beha
2 192.168.1.2:21→192.168.1.3:1402/TCP	Benign	1.000000	The flow shows a balanced request and response traffic with expected TCP handshakes, and reasonable packet counts

Analyzing — may take a minute depending on file size and model speed...

Generate synthetic traffic and analyze it instantly

Packets are built entirely offline with Scapy — nothing is ever sent over a real network — then run through the exact same flow → feature → LLM pipeline as a live capture or upload. Useful for demoing detection without needing live malicious traffic or admin/root privileges.

SYN Flood

~200 SYNs hammering one connection (fixed source/destination) in rapid succession, no ACKs — a single flow with a clean flood signature.

Generate & Analyze

Done

200 synthetic packets → 1 flows classified.

Total flows

1

Benign

0

Suspicious

1

Attack

0

	flow_id	src_ip	classification	confidence	explanation
0	192.0.2.10:80→203.0.113.77:27795/TCP	192.0.2.10	Suspicious	0.800000	The flow has a high SYN count with no ACKs, indicating potential reconnaissance activity. Additionall

Port Scan

One source IP sweeping ~200 destination ports with unanswered SYNs — a reconnaissance signature that only becomes visible across flows sharing a source IP (see the grouped summary after analyzing).

Generate & Analyze

Done

200 synthetic packets → 200 flows classified.

Total flows

200

Benign

164

Suspicious

36

Attack

0

	flow_id	src_ip	classification	confidence	explanation
0	192.0.2.10:65345→198.51.100.77:34276/TCP	192.0.2.10	Benign	1.000000	This flow consists of a single SYN packet with no ACK, PSH, FIN, RST, or URG flags set, indicating a likely initi
1	192.0.2.10:44062→198.51.100.77:11748/TCP	192.0.2.10	Benign	1.000000	The flow has a single packet with balanced request and response traffic, which is consistent with normal cli
2	192.0.2.10:23487→198.51.100.77:43468/TCP	192.0.2.10	Suspicious	0.700000	This flow has a single TCP packet with no ACK, which is unusual, and the SYN flag was sent without an acco
3	192.0.2.10:47326→198.51.100.77:28246/TCP	192.0.2.10	Benign	1.000000	The flow is a single packet with no evidence of scanning, flooding, or denial-of-service, and the protocol fla
4	192.0.2.10:59118→198.51.100.77:56883/TCP	192.0.2.10	Benign	1.000000	The flow has a single packet, balanced request and response traffic, and expected TCP handshake with no f
5	192.0.2.10:2919→198.51.100.77:53259/TCP	192.0.2.10	Benign	1.000000	This flow is classified as Benign due to the very small packet count and byte count, indicating a single, likel
6	192.0.2.10:19947→198.51.100.77:13182/TCP	192.0.2.10	Benign	1.000000	This flow is benign due to a single SYN packet with no responses, indicating a normal client-server handsha
7	192.0.2.10:32747→198.51.100.77:5955/TCP	192.0.2.10	Suspicious	0.800000	The flow exhibits a mild SYN imbalance, with a single SYN packet sent without an ACK response. This unusu

Grouped by source IP — some patterns (like a port scan) only become visible across many flows sharing a source, even though each flow above was classified independently:

	src_ip	flow_count	attack_count	suspicious_count
0	192.0.2.10	200	0	36

Benign Web Browsing

A handful of normal HTTPS-style request/response exchanges with clean handshakes and closes.

Generate & Analyze

Done

30 synthetic packets → 5 flows classified.

Total flows

5

Benign

5

Suspicious

0

Attack

0

	flow_id	src_ip	classification	confidence	explanation
0	192.0.2.50:40000→203.0.113.200:443/TCP	192.0.2.50	Benign	0.950000	This flow is classified as benign due to balanced forward and reverse packet counts (3:3) and a standard TC
1	192.0.2.50:40001→203.0.113.200:443/TCP	192.0.2.50	Benign	0.900000	Normal client-server communication with balanced request and response traffic, indicating a TCP handsh
2	192.0.2.50:40002→203.0.113.200:443/TCP	192.0.2.50	Benign	1.000000	This flow has a balanced request and response traffic with expected TCP handshakes, indicating normal cli
3	192.0.2.50:40003→203.0.113.200:443/TCP	192.0.2.50	Benign	0.950000	The flow has a balanced packet count (3:3 fwd:rev) and no evidence of scanning or flooding attempts. It als
4	192.0.2.50:40004→203.0.113.200:443/TCP	192.0.2.50	Benign	0.950000	This flow exhibits balanced request and response traffic with a reasonable packet count and transfer rate, Ir

Grouped by source IP — some patterns (like a port scan) only become visible across many flows sharing a source, even though each flow above was classified independently:

	src_ip	flow_count	attack_count	suspicious_count
0	192.0.2.50	5	0	0

Live Capture
Upload PCAP
Simulate Attacks
Ask

Ask a question about your flow history

Your question is translated into filters (classification, IP, port, time range) against storage/flows.db — the LLM only ever supplies filter values, never SQL.

Ask something

show me last captured benign results

Ask

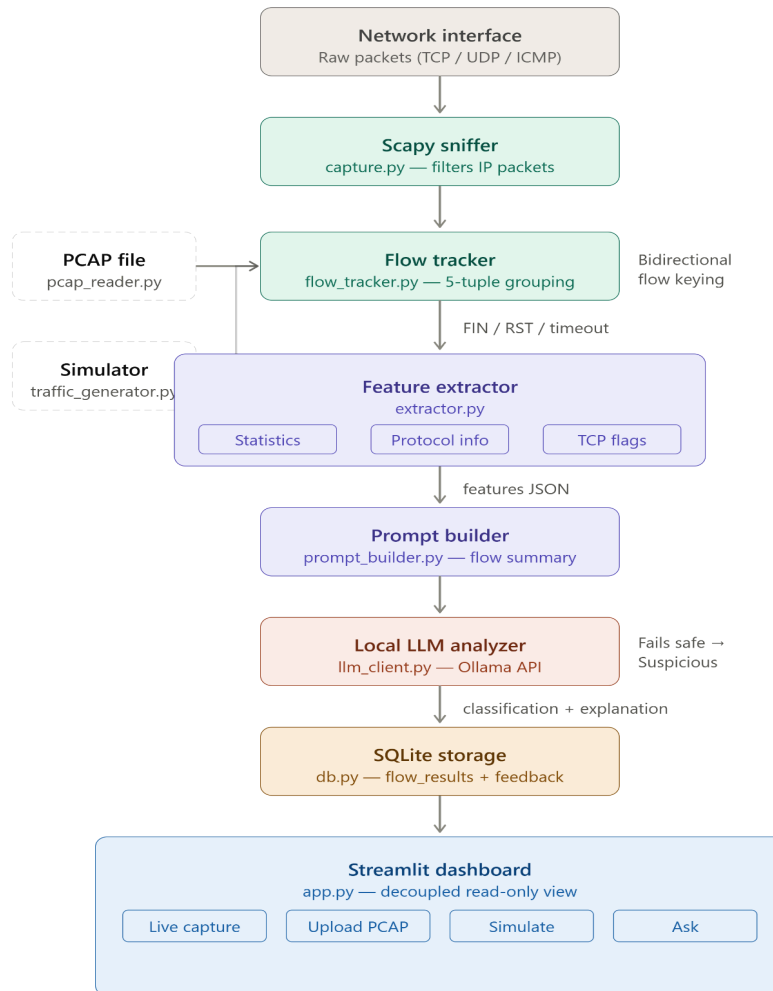
Answer: The most recent benign flows all involve connections to port 443 over either TCP or UDP protocols from various source IPs. For example, one recent flow was initiated by IP address "162.159.61.7" to destination "192.168.10.244:55817" over UDP with a confidence of 0.95.

> Filters this question was translated into

	timestamp	flow_id	classification	confidence	explanation
0	2026-08-02 16:58:50.835888624	192.168.10.244:3769->51.116.246.106:443/TCP	Benign	1.000000	Normal client-server communication with balanced request and response traffic, expected
1	2026-08-02 16:58:49.573556423	192.168.10.244:53139->192.178.134.113:443/UDP	Benign	1.000000	The flow shows balanced forward and reverse packet counts (3 vs 3) with no SYN packets,
2	2026-08-02 16:58:46.892435312	192.168.10.244:3768->51.116.246.106:443/TCP	Benign	0.900000	The flow shows balanced request and response traffic with a reasonable packet count, w/
3	2026-08-02 16:58:45.725664616	192.168.10.244:56631->192.178.158.102:443/UDP	Benign	0.800000	The flow shows a short duration, balanced packet count in both directions, and no signs c

1. **Live Capture:** Allows users to select a network adapter and start/stop live packet capture directly from the browser. It displays real-time results and includes tools for searching flows, generating incident reports, and providing analyst feedback.
2. **Upload PCAP:** Enables the analysis of uploaded *.pcap* or *.pcapng* files through the same pipeline as live capture.
3. **Simulate Attacks:** Generates synthetic traffic scenarios (e.g., SYN flood, port scan) and classifies them instantly. This is particularly useful for demonstrating the system's capabilities without requiring live traffic or administrative privileges.
4. **Ask:** A natural language interface for querying historical flow data, leveraging the LLM to translate questions into database filters and summarize the results.

6. System Architecture Diagram



7. Conclusion

The LLM-IDS project represents a significant step forward in network security monitoring by integrating Large Language Models into the intrusion detection process. Its decoupled architecture, comprehensive dashboard, and fail-safe design make it a robust and user-friendly tool for analyzing network traffic and identifying potential threats. The use of a local LLM (via Ollama) ensures privacy and reduces latency, making it a practical solution for real-time network analysis.